

Web Security Roadmap: Zero to Ultimate

A complete beginner-to-advanced learning plan with famous free resources, timelines, and free certifications.

Phase 0: Prerequisites (Month 0–1)

- Basics of Computers & Internet
- Linux Fundamentals (TryHackMe Pre-Security, Linux Journey)
- Networking Basics (TCP/IP, DNS, HTTP) – Cisco NetAcad, YouTube (freeCodeCamp)

Phase 1: Web Fundamentals (Month 2–3)

- HTML, CSS, JavaScript – freeCodeCamp
- How the Web Works – MDN Web Docs
- HTTP Methods, Cookies, Sessions

Phase 2: Programming for Security (Month 4–5)

- Python for Security – freeCodeCamp
- JavaScript for Security
- Git & GitHub – GitHub Learning Lab

Phase 3: Web Security Basics (Month 6–7)

- OWASP Top 10
- Authentication & Authorization
- Common Attacks: XSS, SQLi, CSRF
- Free labs: PortSwigger Web Security Academy

Phase 4: Hands-on Ethical Hacking (Month 8–9)

- TryHackMe Web Paths
- Burp Suite Basics
- Bug Bounty Basics (HackerOne resources)

Phase 5: Advanced Web Security (Month 10–11)

- Advanced XSS, SSTI, SSRF
- API Security
- OAuth & JWT vulnerabilities
- Cloud Web Security basics

Phase 6: Ultimate Level (Month 12+)

- Bug Bounty Hunting
- CTFs (Hack The Box, TryHackMe)
- Research & Write-ups
- Specialization (Cloud, Mobile Web, Red Team)

Free & Famous Learning Platforms

- freeCodeCamp – Web, JS, Security Basics
- OWASP – Official Documentation
- PortSwigger Web Security Academy – Industry Standard Labs
- TryHackMe – Beginner Friendly Hands-on Labs
- Hack The Box Academy – Advanced Practice
- MDN Web Docs – Web Internals

Free Certifications You Can Earn

- freeCodeCamp Responsive Web Design Certificate
- freeCodeCamp JavaScript Algorithms & Data Structures
- Cisco Networking Basics (NetAcad – Free)
- TryHackMe Certificates (Completion-based)
- PortSwigger Web Security Academy Certificate
- Google Cybersecurity Foundations (Free trial periods)

Daily Time Plan Recommendation

• 1–2 hours/day: Learning & reading • 1 hour/day: Hands-on labs • Weekends: Revision + CTF practice
Consistency for 12 months can make you job-ready or bug-bounty capable.